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“Module 3 Homework”

Load packages

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.5.3
```

```
library(stargazer)
```

```
## Warning: package 'stargazer' was built under R version 3.5.2
```

```
require(aod)
```

```
## Warning: package 'aod' was built under R version 3.5.3
```

```
require(car)
```

```
## Warning: package 'car' was built under R version 3.5.3
```

```
## Warning: package 'carData' was built under R version 3.5.2
```

Q1

```
JTRAIN2 = read.table('JTRAIN2.txt', sep = '', header=FALSE,as.is=TRUE,na.strings = c("NA","."
,""))

colnames(JTRAIN2) = c('train','age','educ','black','hisp','married',
                      'nodegree','mosinex','re74','re75','re78',
                      'unem74','unem75','unem78','lre74',
                      'lre75','lre78','agesq','mostrn')

JTRAIN_NEW=JTRAIN2[,c('train','age','educ','black','hisp','married',
                      'unem74','unem75','unem78')]
```

Q2

```
sum(JTRAIN_NEW$train)
```

```
## [1] 185
```

```
#If a person is participating, then train=1. Hence, 185 people participated.
```

Q3

```
max(JTRAIN2$mostrn)
```

```
## [1] 24
```

```
#24 months is the maximum number a person has participated.
```

Q4

```
#First, let's estimate our models:
#Model 1:
Fit_LPM=lm(unem78~train+age+educ+black+hisp+married+unem74+unem75,
           data = JTRAIN_NEW)
Summary_Fit_LPM=summary(Fit_LPM)

#Model 2:
Fit_Logit=glm(unem78~train+age+educ+black+hisp+married+unem74+unem75,
           data = JTRAIN_NEW, family = binomial(link = "logit"))
Summary_Fit_Logit=summary(Fit_Logit)

#Model 3:
Fit_Probit=glm(unem78~train+age+educ+black+hisp+married+unem74+unem75,
              data = JTRAIN_NEW, family = binomial(link = "probit"))
Summary_Fit_Probit=summary(Fit_Probit)
```

```
#Analysing Train and and EDUC
stargazer(Fit_LPM, Fit_Logit, Fit_Probit, title="Regression Results", align=TRUE,type="html")
```

Regression Results

	Dependent variable:		
	unem78		
	OLS	*logistic*	*probit*
	(1)	(2)	(3)
train	-0.112**	-0.553**	-0.337**
	(0.044)	(0.220)	(0.132)
age	0.00004	0.0004	0.001
	(0.003)	(0.015)	(0.009)
educ	0.0001	-0.002	-0.002
	(0.012)	(0.061)	(0.036)
black	0.189**	1.102**	0.634**
	(0.081)	(0.501)	(0.274)
hisp	-0.038	-0.244	-0.165
	(0.109)	(0.694)	(0.377)
married	-0.025	-0.136	-0.078
	(0.060)	(0.296)	(0.178)
unem74	0.039	0.196	0.106
	(0.072)	(0.352)	(0.212)
unem75	0.016	0.083	0.064
	(0.067)	(0.325)	(0.196)

Constant	0.163	-1.707*	-1.010*
	(0.176)	(0.911)	(0.534)
Observations	445	445	445
R^2	0.046		
Adjusted R^2	0.029		
Log Likelihood		-263.422	-263.313
Akaike Inf. Crit.		544.843	544.626
Residual Std. Error	0.455 (df = 436)		
F Statistic	2.640*** (df = 8; 436)		
Note:	*p<0.1;* ***p<0.05;*** p<0.01		

1. Train variable: The LPM estimate of b1 is equal to -0.112. Hence, participating in the job training program decreases the probability of being unemployed in all of 1978 by 11.2%, AEBE. The Logit model results state that a person who participated in job training, "Log Odds" of being unemployed for 1978 decrease by 0.553. Or, in other words, being unempolyed in 1978 is exp(-0.553)=0.5752216 times less likely if a person participated in job training. The Probit model results state that participation in job training decreases the z-score of P(UNEM78=1|Train,…UNEM75) by 0.337. All 3 b1 estimates in 3 models are statistically significant. No, it does not make sense to compare the coefficients between LPM and Logit/Probit models, because Logit/Probit model coefficients are interpreted differently, and are not really useful by themselves.

2. Educ Variable: The LPM estimate of b1 is equal to 0.0001. Hence, having an extra year of education increases the probability of being unemployed in all of 1978 by 0.01%, AEBE. The Logit model results state that a person who has an extra year of education, "Log Odds" of being unemployed for 1978 decrease by 0.002. Or, in other words, being unempolyed in 1978 is exp(-0.002)=0.998002 times less likely if a person participated in job training. The Probit model results state that participation in job training decreases the z-score of P(UNEM78=1|Train,…UNEM75) by 0.002. All 3 b3 estimates in 3 models are statistically insignificant. No, it does not make sense to compare the coefficients between LPM and Logit/Probit models, because Logit/Probit model coefficients are interpreted differently, and are not really useful by themselves.

Q5

Train, discrete variable:

```
#Approach 1, PEA for Logit:
Means_train_eq1 = data.frame(train=1, age=mean(JTRAIN_NEW$age),
                   educ=mean(JTRAIN_NEW$educ),black=mean(JTRAIN_NEW$black),
                   hisp=mean(JTRAIN_NEW$hisp),
                   married=mean(JTRAIN_NEW$married),
                   unem74=mean(JTRAIN_NEW$unem74),
                   unem75=mean(JTRAIN_NEW$unem75))

Means_train_eq0 = data.frame(train=0, age=mean(JTRAIN_NEW$age),
                             educ=mean(JTRAIN_NEW$educ),black=mean(JTRAIN_NEW$black),
                             hisp=mean(JTRAIN_NEW$hisp),
                             married=mean(JTRAIN_NEW$married),
                             unem74=mean(JTRAIN_NEW$unem74),
                             unem75=mean(JTRAIN_NEW$unem75))

CDF_Logit_train_eq_1=plogis(predict(Fit_Logit, Means_train_eq1), location = 0,
                            scale = 1)
CDF_Logit_train_eq_1
```

```
##         1
## 0.2322916
```

```
CDF_Logit_train_eq_0=plogis(predict(Fit_Logit, Means_train_eq0), location = 0,
                             scale = 1)
CDF_Logit_train_eq_0
```

```
##         1
## 0.3447366
```

```
PEA_Logit=CDF_Logit_train_eq_1-CDF_Logit_train_eq_0
PEA_Logit
```

```
##         1
## -0.112445
```

```
# AEBE, a change in Train from 0 to 1, decreases the probability that an
# average person is unemployed for 1978 by 0.112445.

#Approach 1, PEA for Probit:
CDF_Probit_train_eq_1=pnorm(predict(Fit_Probit, Means_train_eq1), mean = 0,
                             sd = 1)
CDF_Probit_train_eq_1
```

```
##         1
## 0.2329114
```

```
CDF_Probit_train_eq_0=pnorm(predict(Fit_Probit, Means_train_eq0), mean = 0,
                             sd = 1)
CDF_Probit_train_eq_0
```

```
##         1
## 0.3472688
```

```
PEA_Probit=CDF_Probit_train_eq_1-CDF_Probit_train_eq_0
PEA_Probit
```

```
##          1
## -0.1143574
```

```
# AEBE, a change in Train from 0 to 1, decreases the probability that an
# average person is unemployed for 1978 by 0.1143574.

#Approach 2, APE for Logit:
Data_train_eq1 = data.frame(train=1, age=JTRAIN_NEW$age,
                            educ=JTRAIN_NEW$educ,black=JTRAIN_NEW$black,
                            hisp=JTRAIN_NEW$hisp,
                            married=JTRAIN_NEW$married,
                            unem74=JTRAIN_NEW$unem74,
                            unem75=JTRAIN_NEW$unem75)

Data_train_eq0 = data.frame(train=0, age=JTRAIN_NEW$age,
                            educ=JTRAIN_NEW$educ,black=JTRAIN_NEW$black,
                            hisp=JTRAIN_NEW$hisp,
                            married=JTRAIN_NEW$married,
                            unem74=JTRAIN_NEW$unem74,
                            unem75=JTRAIN_NEW$unem75)


MeanCDF_Logit_train_eq_1=mean(plogis(predict(Fit_Logit, Data_train_eq1),
                                     location = 0,scale = 1))
MeanCDF_Logit_train_eq_1
```

```
## [1] 0.2430074
```

```
MeanCDF_Logit_train_eq_0=mean(plogis(predict(Fit_Logit, Data_train_eq0),
                                     location = 0,scale = 1))
MeanCDF_Logit_train_eq_0
```

```
## [1] 0.354245
```

```
APE_Logit=MeanCDF_Logit_train_eq_1-MeanCDF_Logit_train_eq_0
APE_Logit
```

```
## [1] -0.1112376
```

```
#AEBE, on average, a change in Train from 0 to 1, decreases the probability
#that a person will be unemployed in 1978 by 0.1112376.

#Approach 2, APE for Probit:

MeanCDF_Probit_train_eq_1=mean(pnorm(predict(Fit_Probit, Data_train_eq1),
                                     mean = 0,sd = 1))
MeanCDF_Probit_train_eq_1
```

```
## [1] 0.2422247
```

```
MeanCDF_Probit_train_eq_0=mean(pnorm(predict(Fit_Probit, Data_train_eq0),
                                     mean = 0,sd = 1))
MeanCDF_Probit_train_eq_0
```

```
## [1] 0.3545548
```

```
APE_Probit=MeanCDF_Probit_train_eq_1-MeanCDF_Probit_train_eq_0
APE_Probit
```

```
## [1] -0.1123301
```

```
#AEBE, on average, a change in Train from 0 to 1, decreases the probability
#that a person will be unemployed in 1978 by 0.1123301.
```

EDUC, continuous variable:

```
#Approach 1, PEA for Logit:
Means_cv = data.frame(train=mean(JTRAIN_NEW$train),
                      age=mean(JTRAIN_NEW$age),
                      educ=mean(JTRAIN_NEW$educ),
                      black=mean(JTRAIN_NEW$black),
                      hisp=mean(JTRAIN_NEW$hisp),
                      married=mean(JTRAIN_NEW$married),
                      unem74=mean(JTRAIN_NEW$unem74),
                      unem75=mean(JTRAIN_NEW$unem75))

ScaleFactor_Logit_PEA=dlogis(predict(Fit_Logit, Means_cv), location=0, scale=1)

ScaleFactor_Logit_PEA
```

```
##         1
## 0.2078897
```

```
PEA_Logit_CV=ScaleFactor_Logit_PEA*Fit_Logit$coefficients["educ"]

PEA_Logit_CV
```

```
##             1
## -0.0003290766
```

```
#AEBE, an extra year of education, decreases the probability that average
#person will be unemployed in 1978 by approx. 0.00033.

#Approach 1, PEA for Probit:
ScaleFactor_Probit_PEA=dnorm(predict(Fit_Probit, Means_cv), mean=0, sd=1)

ScaleFactor_Probit_PEA
```

```
##         1
## 0.3461828
```

```
PEA_Probit_CV=ScaleFactor_Probit_PEA*Fit_Probit$coefficients["educ"]

PEA_Probit_CV
```

```
##             1
## -0.0006544553
```

```
#AEBE, an extra year of education, decreases the probability that average
#person will be unemployed in 1978 by approx. 0.000654.

#Approach 2, APE for Logit:
ScaleFactor_Logit_APE=mean(dlogis(predict(Fit_Logit), location=0,
                                  scale=1))

ScaleFactor_Logit_APE
```

```
## [1] 0.2030699
```

```
APE_Logit_CV=ScaleFactor_Logit_APE*Fit_Logit$coefficients["educ"]

APE_Logit_CV
```

```
##          educ
## -0.0003214472
```

```
#AEBE, on average, an extra year of education, decreases the probability of
#being unemployed for 1978 by 0.00032.

#Approach 2, APE for Probit:
ScaleFactor_Probit_APE=mean(dnorm(predict(Fit_Probit), mean=0,
                                   sd=1))

ScaleFactor_Probit_APE
```

```
## [1] 0.3362354
```

```
APE_Probit_CV=ScaleFactor_Probit_APE*Fit_Probit$coefficients["educ"]

APE_Probit_CV
```

```
##          educ
## -0.0006356499
```

```
#AEBE, on average, an extra year of education, decreases the probability of
#being unemployed for 1978 by 0.000635.
```

Does it make sense to compare the LPM coefficient estimate on TRAIN with the PEA of TRAIN? Does it make sense to compare the LPM coefficient estimate on TRAIN with the APE of TRAIN? Yes, it does make sense to compate the LPM coefficient estimate on TRAIN with both APE and PEA of train, because APE and PEA both represent the partial effects of the training, and in the LPM model, the coefficient estimates also represent the partial effect of the variable on the probability of being unemployed (since, it is the same as classical linear model). Hence, we can compare these values.

Q6

```
predict_Logit = predict(Fit_Logit, type="response")
predict_Probit = predict(Fit_Probit, type="response")
predict_LPM=predict(Fit_LPM)


#Logit
predict_Logit_min=quantile(predict_Logit,prob=0,na.rm=T)
predict_Logit_max=quantile(predict_Logit,prob=1,na.rm=T)
predict_Logit5=quantile(predict_Logit,prob=0.05,na.rm=T)
predict_Logit25=quantile(predict_Logit,prob=0.25,na.rm=T)
predict_Logit50=quantile(predict_Logit,prob=0.50,na.rm=T)
predict_Logit75=quantile(predict_Logit,prob=0.75,na.rm=T)
predict_Logit95=quantile(predict_Logit,prob=0.95,na.rm=T)
predicted_Logit=c(predict_Logit5, predict_Logit25, predict_Logit50, predict_Logit75, predict_
Logit95)
predicted_Logit
```

```
##        5%       25%       50%       75%       95%
## 0.1200657 0.2379190 0.2929664 0.4165605 0.4182612
```

```
#Probit
predict_Probit_min=quantile(predict_Probit,prob=0,na.rm=T)
predict_Probit_max=quantile(predict_Probit,prob=1,na.rm=T)
predict_Probit5=quantile(predict_Probit,prob=0.05,na.rm=T)
predict_Probit25=quantile(predict_Probit,prob=0.25,na.rm=T)
predict_Probit50=quantile(predict_Probit,prob=0.50,na.rm=T)
predict_Probit75=quantile(predict_Probit,prob=0.75,na.rm=T)
predict_Probit95=quantile(predict_Probit,prob=0.95,na.rm=T)
predicted_Probit=c(predict_Probit5, predict_Probit25, predict_Probit50, predict_Probit75, pre
dict_Probit95)
predicted_Probit
```

```
##        5%       25%       50%       75%       95%
## 0.1183794 0.2374299 0.2958464 0.4149140 0.4191726
```

```
#LPM
predict_LPM_min=quantile(predict_LPM,prob=0,na.rm=T)
predict_LPM_max=quantile(predict_LPM,prob=1,na.rm=T)
predict_LPM5=quantile(predict_LPM,prob=0.05,na.rm=T)
predict_LPM25=quantile(predict_LPM,prob=0.25,na.rm=T)
predict_LPM50=quantile(predict_LPM,prob=0.50,na.rm=T)
predict_LPM75=quantile(predict_LPM,prob=0.75,na.rm=T)
predict_LPM95=quantile(predict_LPM,prob=0.95,na.rm=T)
predicted_LPM=c(predict_LPM5, predict_LPM25, predict_LPM50, predict_LPM75, predict_LPM95)
predicted_LPM
```

```
##        5%       25%       50%       75%       95% 
## 0.1086932 0.2431119 0.2982270 0.4087035 0.4096216
```

```
summary(predicted_Logit)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##  0.1201  0.2379  0.2930  0.2972  0.4166  0.4183
```

```
summary(predicted_Probit)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##  0.1184  0.2374  0.2958  0.2971  0.4149  0.4192
```

```
summary(predicted_LPM)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##  0.1087  0.2431  0.2982  0.2937  0.4087  0.4096
```

```
#LPM model has the lowest mean, max and min values out of all models. The probability of bein
g unemployed is very low according to all models.
```

Q7

```
NewDataQ7=data.frame(train=1, age=52, educ=16, black=0, hisp=0, married = 1,
                     unem74=1, unem75=1)
predict_LogitQ7 = predict(Fit_Logit, NewDataQ7, type="response")
predict_ProbitQ7 = predict(Fit_Probit, NewDataQ7, type="response")
predict_LMPQ7=predict(Fit_LPM, NewDataQ7)
predict_LogitQ7 # The probability is 0.106759
```

```
##        1 
## 0.106759
```

```
predict_ProbitQ7# The probability is 0.105627
```

```
##         1 
## 0.1056278
```

```
predict_LMPQ7# The probability is 0.08525644
```

```
##          1 
## 0.08525644
```

Q8

```
table(Fit_Probit$fitted.values>0.5, JTRAIN_NEW$unem78)
```

```
##
##         0   1
##   FALSE 308 137
```

```
table(Fit_Logit$fitted.values>0.5, JTRAIN_NEW$unem78)
```

```
##
##         0   1
##   FALSE 308 137
```

```
table(Fit_LPM$fitted.values>0.5, JTRAIN_NEW$unem78)
```

```
##
##         0   1
##   FALSE 308 137
```

```
rsquared_logit=(cor(JTRAIN_NEW$unem78, Fit_Logit$fitted.values,
                    use="complete.obs", method="pearson"))^2

rsquared_probit=(cor(JTRAIN_NEW$unem78, Fit_Probit$fitted.values,
                     use="complete.obs", method="pearson"))^2
rsquared_logit
```

```
## [1] 0.04514709
```

```
rsquared_probit
```

```
## [1] 0.04525928
```

```
#The LPM Model has slightly higher "usual" R^2 than pseudo R^2 for Logit and
#Probit models. Hence, LPM is more accurate.
```

Q9

As we have mentioned before, LPM has slightly better Goodness-of-Fit, hence it is more accurate. Moreover, taking into account that both UNEM78 and TRAIN are dummy variables, LPM coefficient estimates are much easier to interpret. Also, if TRAIN goes from 0 to 1 will not increase the probability of unemployment by more than 1, and will not decrease by less than 0.

Q10

```
linearHypothesis(Fit_LPM, c("married=0", "black=0", "hisp=0"), test="F")
```

```
## Linear hypothesis test
##
## Hypothesis:
## married = 0
## black = 0
## hisp = 0
##
## Model 1: restricted model
## Model 2: unem78 ~ train + age + educ + black + hisp + married + unem74 +
##     unem75
##
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1    439 93.156
## 2    436 90.441  3    2.7146 4.3623 0.004838 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
wald.test(b=Fit_Logit$coefficients, Sigma=vcov(Fit_Logit), Terms=c(5,6,7))
```

```
## Wald test:
## ----------
##
## Chi-squared test:
## X2 = 11.8, df = 3, P(> X2) = 0.0081
```

```
wald.test(b=Fit_Probit$coefficients, Sigma=vcov(Fit_Probit), Terms=c(5,6,7))
```

```
## Wald test:
## ----------
##
## Chi-squared test:
## X2 = 13.4, df = 3, P(> X2) = 0.0038
```

```
#Firstly, the joint F test of LPM model shows the p-value of 0.004838. Hence, we reject the n
ull hypothesis that these variables are jointly insignificant. Hence, we disagree with the st
atement of our classmate. The wald tests for Logit and Probit models lead to same conclusion
s. Hence, we disagree.
```

Q11

```
table(Fit_LPM$fitted.values>mean(JTRAIN_NEW$unem78),JTRAIN_NEW$unem78)
```

```
##
##           0   1
##   FALSE 177  53
##   TRUE  131  84
```

```
#Correctly predicted for UNEM78=1: 84/(84+53)=61.31%
#Correctly predicted for UNEM78=0: 177/(177+131)=57.46%
#Overall correctly predicted=58.65%
table(Fit_Probit$fitted.values>mean(JTRAIN_NEW$unem78),JTRAIN_NEW$unem78)
```

```
##
##          0   1
##   FALSE 177  53
##   TRUE  131  84
```

```
#Correctly predicted for UNEM78=1: 84/(84+53)=61.31%
#Correctly predicted for UNEM78=0: 177/(177+131)=57.46%
#Overall correctly predicted=58.65%
table(Fit_Logit$fitted.values>mean(JTRAIN_NEW$unem78),JTRAIN_NEW$unem78)
```

```
##
##          0   1
##   FALSE 177  53
##   TRUE  131  84
```

```
#Correctly predicted for UNEM78=1: 84/(84+53)=61.31%
#Correctly predicted for UNEM78=0: 177/(177+131)=57.46%
#Overall correctly predicted=58.65%

#This approach performs much better, because at the threshold of 0.5, we had
#0 true values predicted correctly. Now, we have correct results with the mean
#approach, because we our prediction models do not have fitted values greater
#than 0.5.
```